

Project Report: Age Calculator Web Application

Python Essentials - Evaluated Course Project

Cover Page

Project: Python Essentials - Evaluated Course Project

Course: Python Essentials

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1. Introduction

Age Calculator is a Flask-based web application that calculates a person's exact age at a selected reference date. It provides a clear browser workflow, server-side validation, local history, and downloadable reports in four formats.

2. Problem Statement

Manual age subtraction is error-prone because calendar months have different lengths and leap years change the number of days in February. Users need a reliable way to calculate age, inspect related birthday information, review previous calculations, and save a readable result.

3. Objectives

- Accept dates in the user-friendly dd/mm/yyyy format.
- Calculate years, months, days, and elapsed-time totals.
- Predict the user's broad age group using a supervised AIML model.
- Handle month lengths and February 29 birthdays consistently.
- Provide clear validation for invalid or incomplete input.
- Allow recent results to be restored from browser history.
- Export a named report in PDF, Excel, Word, or text format.

4. Functional Requirements

Requirement	Implementation
Enter name and dates	Browser form in index.html
Validate dates	JavaScript validation and Flask request validation
Calculate exact age	calculate() in app.py
Display totals and birthday data	Result cards populated by app.js
Store and restore recent calculations	Browser localStorage
Export reports	PDF, XLSX, DOCX, and TXT adapters
Predict age group	Trained 1-nearest-neighbor classifier

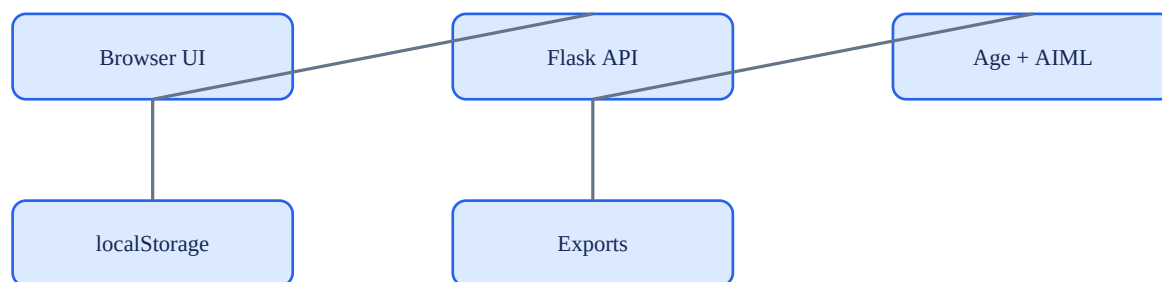
The four major functional modules are calculation, AIML prediction, history, and reporting/export.

5. Non-Functional Requirements

- Usability: Dates are displayed as dd/mm/yyyy, support a calendar picker, and show readable validation messages.
- Reliability: Validation exists in both the browser and API, with automated regression tests.
- Maintainability: Frontend, backend, tests, and documentation are separated into clear files.
- Security: User-entered history values are rendered as text, filenames remove unsafe Windows characters, and malformed requests are rejected.
- Performance: Calculations are in-memory and complete in constant time for normal date inputs.
- Privacy: History remains in browser localStorage and is not persisted on the server.
- Resource efficiency: The classifier uses one numeric feature and a small in-memory training set.

6. System Architecture

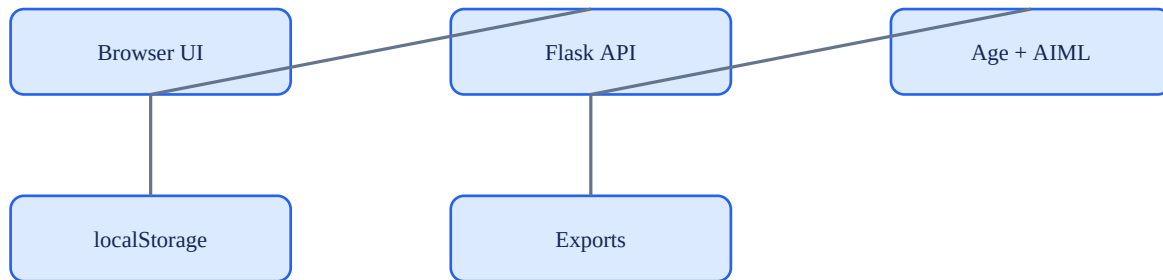
System Architecture



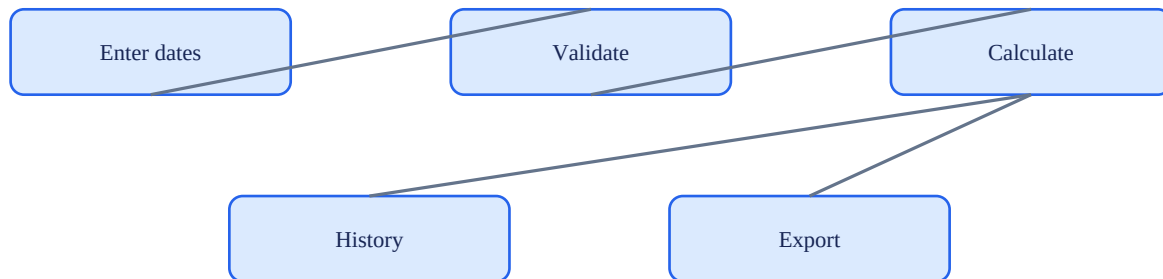
The browser contains the user interface and local history. Flask exposes `/api/calculate` and `/api/export`. The server validates requests, runs date logic, and creates report files. See `architecture.md`.

7. Design Diagrams

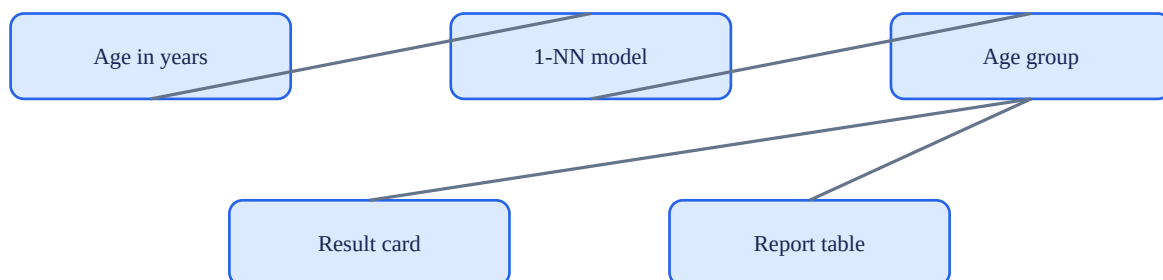
System Architecture



User Workflow



AIML Component / Sequence



The repository includes the required diagrams in `uml.md`, `workflow.md`, and `architecture.md`:

- Use-case diagram
- Component/class diagram
- Sequence diagram
- Workflow diagram
- System architecture diagram
- LocalStorage schema diagram

8. Design Decisions and Rationale

- Flask: Small routing surface and direct integration with Python date logic.
- Browser localStorage: Keeps personal history local without requiring accounts or a database.
- Server-side validation: Prevents clients from bypassing date and export checks.
- On-demand export imports: Export libraries are imported only when the selected format requires them.
- Standard-library unittest: Keeps the test suite easy to run without adding a testing framework dependency.
- Report tables: Tables make PDF and text exports easier to scan than unstructured lines.
- Transparent AIML model: A 1-nearest-neighbor classifier makes the age-group decision easy to explain and audit.
- Version control: Git is used to track source-code, tests, UI, and documentation changes in the project repository.

8.1 AIML Dataset, Model, and Evaluation

The classifier uses a synthetic labeled dataset of integer ages from 0 through 120. Labels are assigned using the project domain definition: 0-12 Child, 13-19 Teenager, 20-59 Adult, and 60 or older Senior. No personal data is used for training.

The selected model is a 1-nearest-neighbor classifier with age as its single feature. This model is appropriate because age-group boundaries are ordered, deterministic, and easy to interpret. The model is trained in memory when the application starts.

Evaluation uses boundary examples and API tests: ages 12, 13, 19, 20, 59, and 60 must map to the correct class. The test suite also verifies the Adult prediction for the sample calculation and ensures the prediction is included in exports.

9. Implementation Details

The browser normalizes typed dates and converts them to ISO format before sending JSON. Flask parses the ISO dates, computes calendar-aware age values, predicts the age group, formats output dates, and returns JSON. Export requests are validated before report generation. The latest ten successful results are stored under age-calculator-history.

10. Screenshots and Results

The running interface contains date inputs, age cards, total-time cards, action controls, and clickable history rows. A typical result includes:

Result	Example
Age	24 years, 10 months, 19 days
AI age group	Adult
Total days	9,089
Total hours	218,136
Total minutes	13,088,160
Report filename	Ankit's age.pdf

Main calculator interface

PERSONAL TIMEKEEPER

AC

Age Calculator

Calculate exact age, total time, and next birthday

Your name OPTIONAL

Enter a name

STARTING POINT

Date of birth

dd/mm/yyyy

REFERENCE DATE

Calculate age on

28/09/2026

Today

YOUR SNAPSHOT

Your age

YEARS

—

MONTHS

—

DAYS

—

AI ML PREDICTION

Estimated age group

Predicted from the calculated age using a trained KNN classifier.

Born on

—

Next birthday

—

Birthday date

—

THE LONG VIEW

Total time

Months

—

Weeks

—

Days

—

Hours

—

Minutes

—

Calculate age

PDF

Save result

Clear

RECENT CHECKS

Calculation history

Clear history

Your completed calculations will appear here.

Main calculator interface

Validation and calculation result

PERSONAL TIMEKEEPER

AC

Age Calculator

Calculate exact age, total time, and next birthday

Your name

OPTIONAL

Mohit

STARTING POINT

Date of birth

31/12/2232

REFERENCE DATE

Calculate age on

28/09/2026

Today

YOUR SNAPSHOT

Your age

YEARS

—

MONTHS

—

DAYS

—

AIML PREDICTION

Estimated age group

Predicted from the calculated age using a trained KNN classifier.

Born on

—

Next birthday

—

Birthday date

—

THE LONG VIEW

Total time

Months

—

Weeks

—

Days

—

Hours

—

Minutes

—

The calculation date cannot be before the date of birth.

Calculate age

PDF

Save result

Clear

RECENT CHECKS

Calculation history

Clear history

Your completed calculations will appear here.

Validation error

PERSONAL TIMEKEEPER

AC

Age Calculator

Calculate exact age, total time, and next birthday

Your name OPTIONAL

Mohit

STARTING POINT

Date of birth

06/08/2006

REFERENCE DATE

Calculate age on

28/09/2026

Today

YOUR SNAPSHOT

Mohit's age

YEARS

20

MONTHS

1

DAYS

22

AIML PREDICTION

Estimated age group

Adult

Predicted from the calculated age using a trained KNN classifier.

Born on

Sunday

Next birthday

312 days

Birthday date

August 06, 2027

THE LONG VIEW

Total time

Months

241

Weeks

1,051

Days

7,358

Hours

176,592

Minutes

10,595,520

Calculate age

Text

Save result

Clear

RECENT CHECKS

Calculation history

Clear history

Mohit

Born: 06/08/2006

On: 28/09/2026

Adult | 20 years, 1 months, 22 days

Calculation result

Calculation history

AC

PERSONAL TIMEKEEPER

Age Calculator

Calculate exact age, total time, and next birthday

Your name

OPTIONAL

Mohit

STARTING POINT

Date of birth

06/08/2006

REFERENCE DATE

Calculate age on

28/09/2026

Today

YOUR SNAPSHOT

Mohit's age

YEARS

20

MONTHS

1

DAYS

22

AIHL PREDICTION

Estimated age group

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Predicted from the calculated age using a trained KNN classifier.

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Save result

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Mohit

Born: 06/08/2006

On: 28/09/2026

Adult | 20 years, 1 months, 22 days

Calculation history

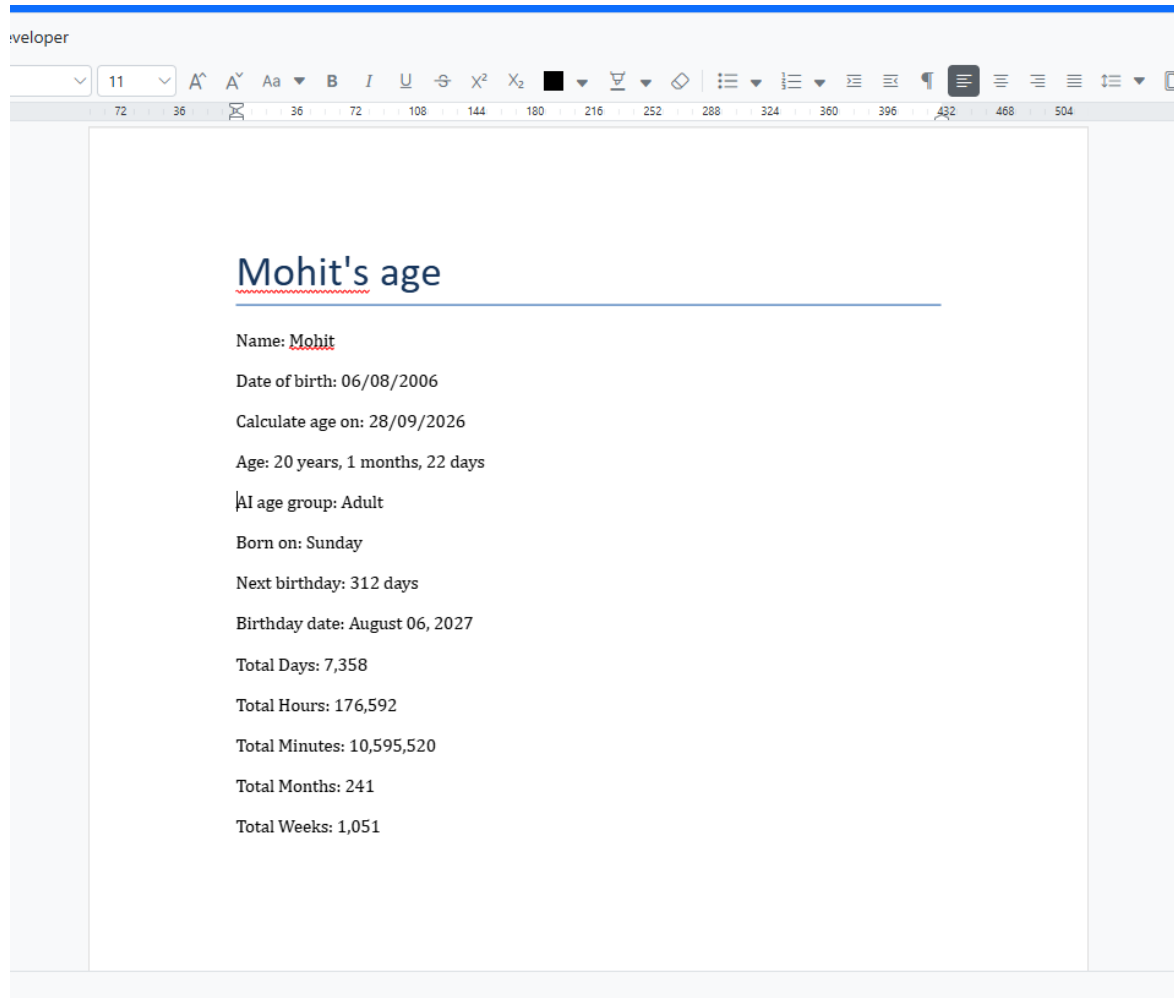
Mohit's age

Details	Result
Name	Mohit
Date of birth	06/08/2006
Calculate age on	28/09/2026
Age	20 years, 1 months, 22 days
AI age group	Adult
Born on	Sunday
Next birthday	312 days
Birthday date	August 06, 2027
Total Days	7,358
Total Hours	176,592
Total Minutes	10,595,520
Total Months	241
Total Weeks	1,051

PDF export

← Mohit's age			
File ▾ Home Insert Formulas Data View			
↶ ↷ ✂ 📄 📄 ▾ General ▾			
N10 ▾ fx			
	A	B	C
1	Details	Result	
2	Name	Mohit	
3	Date of birth	06/08/2006	
4	Calculate age on	28/09/2026	
5	Age	20 years, 1 months, 22 days	
6	AI age group	Adult	
7	Born on	Sunday	
8	Next birthday	312 days	
9	Birthday date	August 06, 2027	
10	Total Days	7,358	
11	Total Hours	176,592	
12	Total Minutes	10,595,520	
13	Total Months	241	
14	Total Weeks	1,051	
15			
16			
17			

Excel export



Word export

MOHIT'S AGE

Details	Result
Name	Mohit
Date of birth	06/08/2006
Calculate age on	28/09/2026
Age	20 years, 1 months, 22 days
AI age group	Adult
Born on	Sunday
Next birthday	312 days
Birthday date	August 06, 2027
Total Days	7,358
Total Hours	176,592
Total Minutes	10,595,520
Total Months	241
Total Weeks	1,051

Text table export

11. Testing Approach

The project uses Python unittest with Flask's test client. The ten tests cover normal calculation, age-group boundary predictions, leap-day birthdays, invalid JSON, invalid dates, reversed dates, text-table output, named filenames, incomplete export data, unsupported formats, and non-numeric export values.

Run:

Latest verification: 10 tests passed.

12. Challenges Faced

- Handling different month lengths without producing negative day values.
- Defining a consistent February 29 birthday rule in non-leap years.
- Keeping the browser's dd/mm/yyyy display separate from the API's ISO format.
- Producing readable table layouts across PDF and plain-text exports.
- Keeping filenames useful while removing characters invalid on Windows.
- Selecting an AIML approach that is explainable and small enough to run without a model service.

13. Learnings and Key Takeaways

- Client-side validation improves usability, but server-side validation is still required for correctness.
- Date calculations need explicit rules for leap years and month boundaries.
- A small application benefits from separate UI, API, calculation, export, test, and documentation responsibilities.
- Automated tests expose edge cases that normal examples do not reveal.
- Simple supervised models can add useful domain classification without collecting a personal dataset.

14. Future Enhancements

- Add optional database-backed, cross-device history.
- Add localization for date formats and interface language.
- Add browser end-to-end tests and automated screenshot capture.
- Add report preview and configurable report templates.
- Add user accounts only if cloud history becomes necessary.
- Expand the age-group model with additional features only when a suitable consented dataset is available.

15. References

- Python datetime and calendar standard-library documentation.
- Flask documentation for routing, JSON requests, and test clients.
- ReportLab documentation for PDF tables.
- openpyxl documentation for workbook creation.
- python-docx documentation for Word document creation.
- Mermaid documentation for Markdown diagrams.